MTH 200 Chapter 3: Derivatives

3.5 Trig. Derivatives (requires 3.3)

****Remember that simplifying before attempting to find the derivative is worthwhile****

1. Differentiate the following functions and fully simplify.

a) $g(t)=\dfrac{t^3}{\sec(t)}+\csc(t)$

b) $y=\sec(x)\cot(x)\sin(x)\cos(x)\csc(x)$

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c) $h(x)=e^x\sin(x)\tan(x)$

d) $f(x)=\dfrac{\sin(x)}{1+\cos(x)}$

2. Use your knowledge that $\dfrac{d}{dx}\left[\sin(x)\right]=\cos(x)$ to show that $\dfrac{d}{dx}\left[\csc(x)\right]=-\csc(x)\cot(x)$.

3. If $H(x) = x \cdot \sec(x)$, find $H'(x)$ and $H''(x)$

4. Find the second derivative $f''(t)$ of the function $f(t) = \csc(t)$.

5. Suppose $f\left(\frac{\pi}{3}\right) = 4$ and $f'\left(\frac{\pi}{3}\right) = -2$, and let $g(x) = f(x) \cdot \sin(x)$ and $h(x) = \frac{\cos(x)}{f(x)}$.

a) Find $g'\left(\frac{\pi}{3}\right)$.

b) Find $h'\left(\frac{\pi}{3}\right)$.

6. What is the equation of line tangent to the graph of $y = 2\tan(x) + 3x^2 + x + 7$ at its *y*-intercept?

7. For what values of *x* does the graph of $f(x) = x + 2\sin(x)$ have a horizontal tangent?